ResearchGate

See discussions, stats, and author profiles for this publication at: https://www.researchgate.net/publication/308779493

Understanding Personality Development: An Integrative State Process Model

Article *in* International Journal of Behavioral Development · December 2016

DOI: 10.1177/0165025416677847

CITATIONS
101

READS
11,422

3 authors:

Katharina Utesch
German Aerospace Center (DLR)
102 PUBLICATIONS **2,359** CITATIONS

SEE PROFILE

Maarten Van Zalk
Osnabrück University
92 PUBLICATIONS **4,271** CITATIONS

SEE PROFILE

Mitja D Back
University of Münster
305 PUBLICATIONS **12,083** CITATIONS

SEE PROFILE

All content following this page was uploaded by Maarten Van Zalk on 14 February 2019.

The user has requested enhancement of the downloaded file.

Check for updates

ijbd

International Journal of Behavioral Development
2018, Vol. 42(1) 43–51
© The Author(s) 2017
Reprints and permissions:
sagepub.co.uk/journalsPermissions.nav
DOI: 10.1177/0165025416677847
journals.sagepub.com/home/ijbd
SAGE

Understanding personality development: An integrative state process model

Katharina Geukes, Maarten van Zalk, and Mitja D. Back

Abstract

While personality is relatively stable over time, it is also subject to change across the entire lifespan. On a macro-analytical level, empirical research has identified patterns of normative and differential development that are affected by biological and environmental factors, specific life events, and social role investments. On a micro-analytical level, however, little is known about the underlying processes driving personality development. We provide an integrative state process model of personality development that incorporates and builds on previous process approaches to personality. It integrates this micro-analytical state perspective into the well-established macro-analytical understanding of personality trait development. Specifically, we distinguish three domains of individual differences in (the level of and contingencies between) state processes: (1) Goals and Strategies, (2) Actions and Experiences, and (3) Evaluations and Reflections. These process domains build a continuous sequence, with each domain guiding state processes in the next. Each process domain itself and their dependencies within the succession may be subject to change, thereby reflecting normative (in the case of shared development in state processes) and/or differential trait development (in the case of unique development in state processes). Well-established effects of environmental and biological structures, social roles, age, and life events on personality trait development can be explained by systematic links of these macro-level determinants to the outlined micro-level state processes. This integrative, process-based approach is thought to provide a conceptual basis for empirical research aiming at a comprehensive and fine-grained process understanding of personality development across the lifespan.

Keywords

behaviors, goals, if-then contingencies, goals, life events, personality processes, social information-processing, social relationships, social roles, state processes

Empirical research on personality development has provided considerable evidence for personality development across the entire lifespan—including global personality traits within the Big Five framework (e.g. Roberts, Donnellan, & Hill, 2012) and self-esteem (e.g. Orth & Robins, 2014; Robins & Trzesniewski, 2005). On a macro-analytical level, empirical studies have identified personality development to be determined by genetic and environmental factors (for recent overviews, see Bleidorn, Kandler, & Caspi, 2014; Specht et al., 2014) as well as by more specific factors, such as life events (e.g. Lüdtke, Roberts, Trautwein, & Nagy, 2011; Specht, Egloff, & Schmukle, 2011) and social role investments (e.g. Denissen, Ulfers, Lüdtke, Muck, & Gerstorf, 2014; Lodi-Smith & Roberts, 2007). On a micro-analytical level, however, empirical insights into the processes underlying these developments are still limited. Therefore, and adding to recent process-oriented models of personality development, such as the TESSERA model (Wrzus & Roberts, 2016), the goal of this article is to provide a generic state process model of personality development that integrates a micro-level state-process perspective on personality into the macro-level understanding of personality development.

Approaching processes in personality development

On the most general level, our model (see Figure 1) builds on previous research concerning environmental and biological factors as distal determinants of personality development (for overviews, see Roberts, Wood, & Caspi, 2008; Specht et al., 2014), including the Social Investment Principle (Lodi-Smith & Roberts, 2007) and the Neo-Socioanalytical Model (Roberts & Wood, 2006; see also Roberts, 2006). We incorporate biological structures (e.g. gene expression, hormonal levels, physical abilities), environmental structures (e.g. probabilities of certain situational contexts, types of interaction partners), social roles (e.g. being a mother, friend, employee), age, and life events (e.g. marriage and divorce, birth of a child, accidents) as important macro-level factors affecting personality development.

Into the center of this macro-level framework, we place a micro-level, process-oriented perspective on personality. The sociogenomic model (Roberts & Jackson, 2008; see also Roberts, 2009) was the first to explicitly highlight the key role of states as driving trait change by means of bottom-up processes. This idea has been expanded and related to previous theoretical and empirical insights in the recently published TESSERA model (Wrzus & Roberts, 2016). We built on this groundbreaking work and provide a fresh perspective on how this and other perspectives on personality development might be further strengthened. Specifically, we

University of Münster, Münster, Germany

Corresponding author:
Katharina Geukes, University of Münster, Fliednerstr. 21, Münster, 48149, Germany.
Email: katharina.geukes@wwu.de

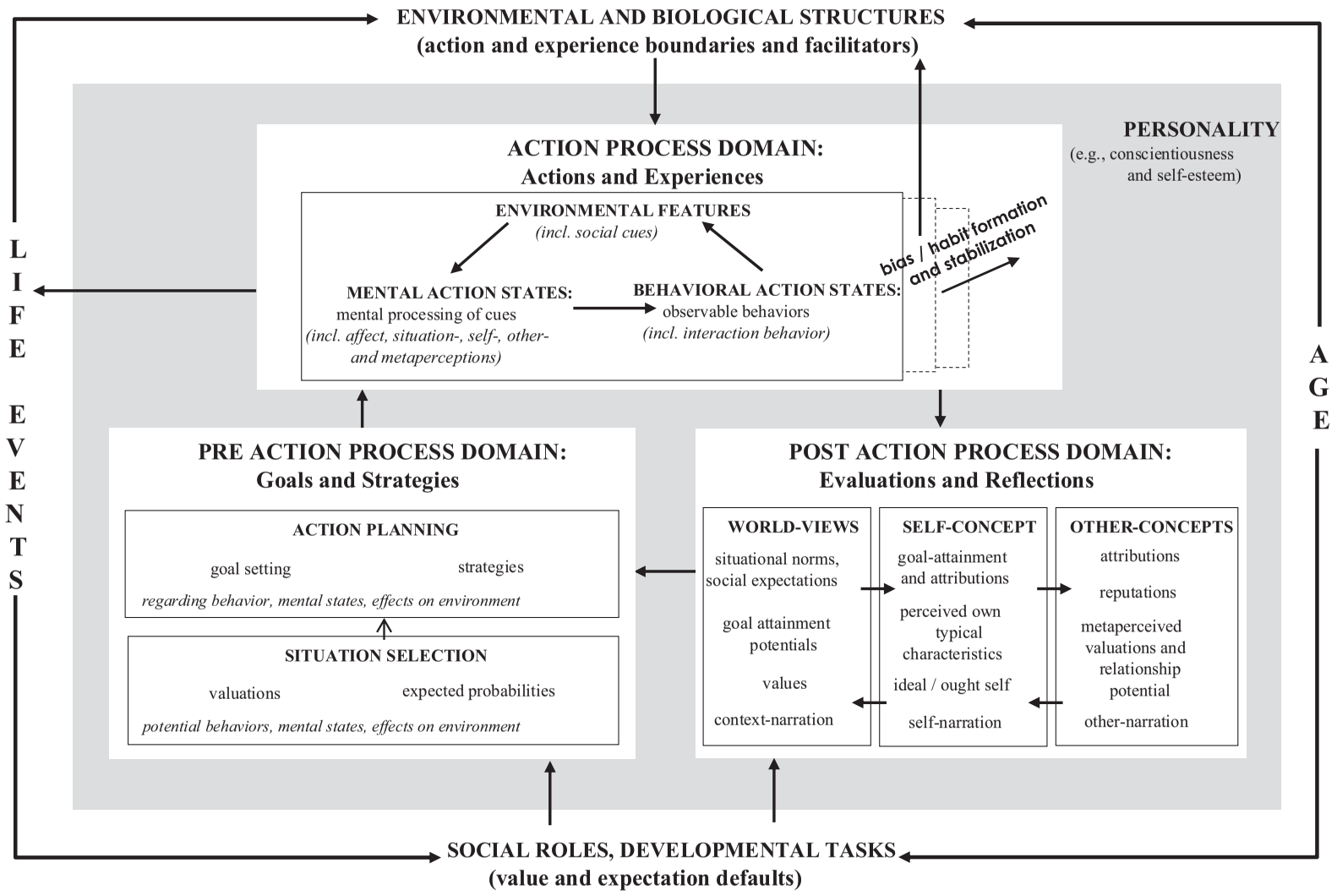

Figure 1. A generic, integrative state process model of personality development.

present a unique integration of three theoretical ingredients that have not previously been combined, and that we regard as important und useful for fostering a process-based perspective on personality development.

First, following the TESSERA framework (Wrzus & Roberts, 2016) and the PERSOC framework (Back et al., 2011), we assume that long-term personality trait development is driven by changes regarding recursive sequences of specific and systematically connected state processes. In the TESSERA framework, these process sequences are described as TESSERA sequences including Triggering situations (e.g. psychologically relevant environmental features of a situation), Expectancies (e.g. intentions/goals triggered by the situation), States/State Expressions (momentary thoughts, feelings, and behaviors), and ReActions (e.g. reinforcement by oneself or others). The PERSOC framework (Back et al., 2011) focuses on the interplay of personality and social relationships, and process sequences are described as social interaction units made up of interconnected interaction partners' behavioral (i.e., interaction behaviors) and mental states (including self-, other-, and metaperceptions).

Second, we adopt the general behavior-/self-regulatory cycle (Carver & Scheier, 1981; Gollwitzer, 1990; Gross & Thompson, 2007; Heckhausen, 1991) to sort state processes into pre-action, action, and post-action domains. In doing so, we use state processes that have been described in detail within functional self-regulatory perspectives on personality development (Denissen, Wood, Penke, & van Aken, 2013; Hennecke, Bleidorn, Denissen, & Wood, 2014; Wood & Denissen, 2015), including changes in typical behaviors, evoked by changes in expectations, reference values, personal goals, and social norms.

Third, we follow a state perspective on personality traits that deconstruct the idea of latent traits and conceptualizes traits as being networks of connected states within individuals in search of equilibrium. This individual network approach (Bringmann et al., 2016; Cramer et al. 2012; Schmittmann et al., 2013) is similar to other process-oriented models of personality, in that traits are defined as individual differences in *state processes*—the dynamic patterns of how people feel, think, strive for, and behave in specific situations (e.g. Cervone, 2005; Denissen & Penke, 2008; Fleeson & Jayawickreme, 2015; Mischel & Shoda, 1998; Robinson, 2007, Robinson & Wilkowski, 2015; Wood, Gardner, & Harms, 2015; Wright & Hopwood, 2016). Within the individual network approach, personality traits are understood as a within-person system of interconnected cognitive, affective, and behavioral state components that mutually reinforce each other for causal, homeostatic, or logical reasons—without the assumption of an additional "common cause" of all state variables, that is, without a latent trait variable. Individual differences regarding these individual networks pertain to the *level of state expressions* (i.e., the activity of network nodes: individual differences in how much individuals typically express a state variable; e.g. how strong an angry feeling is) and to *contingencies of state expressions* (i.e., the strength of network ties: individual differences in how much state variables depend on each other; e.g. how much perceived negative feedback relates to angriness). The strong temporal stability of globally assessed personality traits is a consequence of these states tending to develop into stable configurations (i.e., state network equilibria) within individuals. Accordingly, long-term personality development is suggested to occur when components of this system (levels and/or contingencies of different state expressions) deviate from this balanced configuration, leading to systemic change of the interconnected components until a new homeostatic equilibrium is found.

While all of these three theoretical ingredients have been outlined in detail (see Cramer et al., 2012; Wood & Denissen, 2015; Wrzus & Roberts, 2016), they have not yet been integrated into one single model. The goal of the present article is to present a first brief and generic idea of what such an integration might look like and to point at some conceptual and methodological implications of this unique integrative perspective. We will now describe our process model in further detail and therefore introduce Anna as an example. Anna is a 32-year-old lawyer, and she has recently become a mother. Anna can be characterized as being high in conscientiousness and low in self-esteem.

Three domains of state processes

Following a process perspective of personality, individual differences in personality traits can be understood as relatively enduring individual differences in the levels and contingencies of state expressions. Here, we distinguish three broad domains of state processes that each refer to a distinct phase of behavior regulation and reflect different research traditions regarding the definition and assessment of personality (see Figure 1): (1) *Goals and Strategies* refer to pre-action phases and reflect individual differences in motives, (2) *Actions and Experiences* refer to action phases and reflect individual differences in behaviors and cognitive-affective processing units, and (3) *Evaluations and Reflections* refer to post-action phases and reflect individual differences in self-concepts/identity narratives, reputations, general attitudes and values.[1]

Goals and strategies

The domain of *Goals and Strategies* concerns individual differences in processes of (1) selection of situations and (2) creation of action plans for a given situational choice (see Wood & Denissen, 2015). Situation selection depends on three interrelated estimations: the valuations of desired outcomes (i.e., motives, desired end-states, reference values), the expectations regarding one's abilities of performing required actions, and the anticipated effectiveness of these actions leading to desired outcomes (see Feather, 1959; Hastie, 2001; Rotter, 1954). Thus, individuals would select those situations that they assume to result in desired outcomes and that involve actions they expect to be able to perform and to have the desired effects. Once a situation is selected, individuals create according action plans that involve setting goals and selecting accompanying strategies to achieve these goals. Thus, individuals would change in their personality to the extent they change in (1) situation selection (i.e., situational valuations, situational expectations, evaluations of the effectiveness of actions in given situations and their competence to perform these actions), and/or (2) action plan creation (i.e., goal setting; strategy selection; e.g. novel action plans that fit these adjusted valuations and expectations; e.g. Denissen et al., 2013; Hennecke et al., 2014).

Imagine highly-conscientious Anna. She aims at achieving her life's ambition of becoming a lawyer, and typically selects achievement situations over social situations. She would, for instance, typically prefer to keep on studying on a Saturday night, rather than to go out with friends. This preference is based on (a) her valuation of her desired outcome, (b) her conviction that she is able to work hard, and (c) her expectation that, when working hard enough, she will succeed at entering law school and become a lawyer. Based on the selection of working situation, Anna engages in action planning,

that includes the ambitious goal to complete more tasks than do all of her colleagues, and more than she has last week, and the rigid strategy to work as long as she is able to stay awake. To change her conscientiousness, Anna might, for example, need to change her valuation of becoming a lawyer (e.g. value social inclusion more) and/or to adjust her overly ambitious goals (e.g. aim for half of the task completion), and/or rigid strategies (e.g. allow herself to take regular breaks during working hours and socializing on weekends).

Actions and experiences

The domain *Actions and Experiences* includes individual differences regarding three mutually dependent components: (1) environmental features of the selected situation (including cues of social partners), (2) mental action states (e.g. affect, cognitions, evaluations, social perceptions), and (3) behavioral action states (i.e., observable behaviors influenced by the mental processing and that, in turn, shape environmental features). Central to this process domain is the interdependence between actually experienced and expressed states, their environmental precursors, and their direct consequences.

Following this reasoning, personality change would result (a) if mental state levels change (e.g. perceiving yourself or others to be likeable; experiencing positive affect) and behavioral state levels change (e.g. smiling, working hard), and/or (b) if contingencies between environmental features and mental processing change (e.g. how much positive feedback leads to positive self-perception; how much an intellectually challenging task leads to hard working) and if contingencies between mental and behavioral states change (e.g. how much positive self-perceptions lead to smiling; how much positive affect undermines or fosters hard working). Joint repetitions of mental and behavioral states are assumed to lead to greater automatization and independence of environmental cues and effects. This way, mental biases stabilize, behavioral habits form, or, more generally, learning takes place.

Anna tends to perceive situations as achievement-related, even social situations (e.g. playing charades with friends) that others rather perceive as companionable fun. This leads her to experience negative affect and to worry about her "performance". Therefore, she rather behaves in a controlled but agitated fashion, being unable to hide her nervousness that typically goes along with rather negative social feedback. To change her self-esteem, for example, the interlinkage of negative social feedback, negative affective mental states, and nervous behaviors would need to be interrupted. One day, Anna might be in a good mood because she received the desired scholarship. When playing charades with her friends in the evening, she does not worry as much about her "performance" and she creatively and entertainingly masters all the charades-related and social challenges. As a response to this behavior, she receives positive social feedback. The reduced level of negative affectivity, thus, results in more self-assured behavior that evokes more positive social feedback and might start a positive spiral. However, those incidents would need to occur repeatedly so that positive affectivity and self-confident behavior become increasingly automatized and habitual for Anna.

Evaluations and reflections

The domain of *Evaluations and Reflections* concerns individual differences in processes regarding (1) individual's self-concepts, (2) their concepts of other individuals (i.e., the generalized other) and others' concept of individuals (i.e., reputations), and (3) their world-views (i.e., their concepts about environmental contexts and circumscribed situations within these contexts). It is characterized by reflective processes (e.g. life reflection, self-narration) regarding individuals' selves and their social counterparts, the environments, and the world they live in (also see Caspi & Roberts, 1999). This includes, for example, processes related to self-narration (Bauer & McAdams, 2004; Dunlop, 2015; Lodi-Smith, Geise, Roberts, & Robins, 2009; Pasupathi, Mansour, & Brubaker, 2007) and the social structure of selves (e.g. actual vs. ideal vs. ought selves; Higgins, 1987). Changes in Evaluations and Reflections in one of the three subgroups (i.e., world-views, self-concept, other-concepts), or in their interdependencies, are thought to contribute to permanent trait change.

Anna might attribute her failures (e.g. a low grade in the criminal law exam, an unsuccessful job application in a corporate law firm) to not having worked hard enough. And although she recognizes others to be happier and more confident while being far less successful than she is, she still evaluates their approach to life as irresponsible and hedonistic given the competitive world we live in. To change her Evaluations and Reflections, Anna would need to alter her constructions of the self, others, and the world towards a more conciliatory view of herself (to become more independent from her academic or professional achievements), a more realistic view of others (who perhaps are less ambitious but not necessarily hedonic), and the world (some law firms may be more competitive than others).

The interplay of state processes

Within each process domain, the ease and effectiveness by which a given state change leads to personality trait change depends on its strength and frequency and the degree of its connectedness to other state processes (i.e., the centrality of a state in an individual network: Cramer et al., 2012; see also research on the self-reinforcing nature of social information processing mechanisms: Crick & Dodge; 1994; Nickerson, 1998). Accordingly, changes in one state might either lead to permanent change as it pushes other states to similar levels or, alternatively, it might lead to momentary fluctuations as the other states might pull the changed state back to its previous state level, and to either change or stability of personality, accordingly. Importantly, across process domains, respective domain-specific processes are not independent of each other, but form a sequence of temporarily and logically depending consecutive processes. Here, *Goals and Strategies* inform individual's mental and behavioral states when entering an action unit—that is an individuals' *Actions and Experiences*. Subsequently, these Actions and Experiences (i.e., own and others' behaviors, features of the environment, and how these were influenced by behaviors) guide *Evaluations and Reflections* (i.e., about oneself, others, and the world). Finally, these Evaluations and Reflections provide the basis for situational valuations and expectations, that is, for Goals and Strategies. When it comes to personality development, stabilizations and changes in one domain will, therefore, tend to affect stabilization and changes in other domains.[2]

Understanding macro-level determinants of personality development

Integrating the described process approach into the macro-level understanding of personality development might help to explain

how macro-level determinants—such as biological structures, environmental structures, life events, social roles, and age—lead to stabilization of and/or change in personality (see Figure 1). The causal processes implied in the present model follow the described individual network perspective on personality traits (macro-level determinants cause changes within the dynamic network of states, that subsequently find a new equilibrium, representing trait change; see Cramer et al., 2012), and, thus, differ from mediational logic described in previous models (macro-level determinants causes state changes which causes changes in the latent trait).

Biological structures (e.g. individual differences in gene expression, hormonal levels) can be understood as set points for Actions and Experiences as they inhibit or foster the experience of certain mental states and the expression of certain state behaviors (e.g. Headey & Wearing, 1989; Ormel, Riese, & Rosmalen, 2012). At the same time, according to a bottom-up approach to personality development, enduring changes in experiential and behavioral states can lead to permanent changes in general behavioral and mental tendencies (see Roberts & Jackson, 2008; see bias/habit formation in Figure 1). In the long run, this can affect neuroanatomical structures and/or gene expression, thereby changing biological set-points. Biological changes might additionally be triggered by ageing (e.g. genetically determined set-points are activated at a certain age) and major life events (e.g. effects of injuries, drug consumption, traumatic experiences).

Changes in environmental structures (e.g. culture and context differences in patterns of environmental cues), similarly, might lead to personality change as they determine the range, frequency, and intensity of expressed behavioral and experienced mental states (i.e., socialization effects). These changes can additionally be affected by ageing and life events, with some life events being actively chosen by individuals (i.e., selection effects; Caspi & Roberts, 1999).

Social roles (Caspi, 1987; Roberts & Wood, 2006) and developmental tasks (Erikson, 1950; Havighurst, 1972; Hutteman, Hennecke, Orth, Reitz, & Specht, 2014) directly affect specific state processes in the domains of Evaluations and Reflections (e.g. perceived social norms; comparisons between actual, ideal, and ought selves) and Goals and Strategies (e.g. valuation of situational options; expected ability to perform certain actions). They typically change as individuals get older (i.e., they are age-graded) and are additionally related to life events (e.g. different social norms for individuals with or without children).

Understanding normative and differential patterns of personality development

The outlined process perspective on personality development applies to both the understanding of normative personality development—when the development of state processes is shared within a population, thereby reflecting mean-level changes in a population—and the understanding of differential personality development—when state processes develop in a way that is unique to certain individuals within a population, thereby reflecting rank-order changes, and thus, a lowered relative stability.

Regarding patterns of normative development (mean-level change), previous research has, for example, found increasing mean-levels of conscientiousness (e.g. Bleidorn, 2015; Srivastava, John, Gosling, & Potter, 2003) and self-esteem (e.g. Orth, Trzesniewski, & Robins, 2010; Trzesniewski, Donnellan, & Robins, 2003) from adolescence to adulthood (i.e., personality maturation). Following the outlined process approach, future research should try to unravel the kind of changes in state dynamics that are shared among members of a population and that drive normative personality development. Anna's increased conscientiousness, for instance, might be partially explained by a normative development dynamic. Like other individuals of her age, Anna might have experienced a particular boost in conscientiousness in the final year of college (e.g. Bleidorn, 2012; Wagner, Lüdtke, Jonkmann, & Trautwein, 2013). Following our model, age-related social roles might simultaneously affect many individuals in the transition from school to university and/or working life. Specifically, in response to social roles, students might share anticipated social expectations and ideal-selves (being a productive worker, being a dependable citizen; Evaluation and Reflections) and an enhanced valuation of achievement-related outcomes (Goals and Strategies). For Anna and her peers, this might result in repeated states of hard-working behavior and stronger contingencies between work-related environmental cues and positive affect (Actions and Experiences) and the positive evaluation of such behavior by Anna and by others (Evaluations and Reflections), thereby initiating collective changes in state dynamics towards more conscientious configurations (i.e., normative change in conscientiousness).

Regarding patterns of differential personality development (i.e., rank-order stability/change), previous research has shown both, a substantial stability of conscientiousness (e.g. Roberts, Walton, & Viechtbauer, 2006; Specht, et al., 2011) and self-esteem (e.g. Donnellan, Kenny, Trzesniewski, Lucas, & Conger; 2012; Orth & Robins, 2014; Wagner, Lüdtke, & Trautwein, 2015) as well as differential change based on individual differences in the presence, timing, and experience of life events and social role transitions (e.g. Bleidorn, 2012; Denissen, et al., 2014; Hutteman, Bleidorn, et al., 2014; Hutteman, Nestler, Wagner, Egloff, & Back, 2015; Lüdtke, et al., 2011; Reitz, Motti-Stefanidi, & Asendorpf, 2013; Specht et al., 2011; Wagner, Gerstorf, Hoppmann, & Luszcz, 2013; Zimmermann & Neyer, 2013). The comparatively strong stability of personality (i.e., coherent individual differences) can be aligned to the hypothesized close connections between state processes within and across process domains. The outlined state process dynamics can be applied to explain both the stabilization of individual differences as well as rank-order changes.

Anna's stable low level of self-esteem might result from highly interconnected state processes that are already automatized in their succession. She might habitually devalue social goals and have low expectations regarding her social effectiveness, and set herself overly ambitious academic goals and rigid behavioral strategies (Goals and Strategies). This might influence a reduced reactivity to social cues, anxious reactions to signs of competitiveness and unresting working habits, which undermine positive social reactions yet produce good academic outcomes that are, nevertheless, below her aspired standards (Actions and Experiences). This, in turn, translates into metaperceived social disapproval and into negative evaluations of goal attainment, both of which strengthen her achievement-oriented and socially isolated, competitive worldviews, and a negative self-concept below her achievement ideals (Evaluations and Reflections), which further foster her motivational set-up (Goals and Strategies).

Unique changes in state processes (e.g. unique state reactions to certain life events or role transitions) might underlie Anna's differential personality development, i.e., why she developed high conscientiousness and low self-esteem. Anna's recent motherhood

might have resulted in a much smaller decrease or even increase in her conscientiousness (see Hutteman, Bleidorn et al., 2014) as compared to others, as she has quickly adopted to the new role and expectations, has re-defined her goals as a working mum (Goals and Strategies), has found her personal strict parenting style while perceiving childcare as an achievement-related task (Actions and Experiences), and has correspondingly further adopted an achievement-focused self-concept, other-concept, and world-view (Evaluations and Reflections).

As another example, Anna might have experienced stronger decreases in self-esteem than other mothers each time when giving birth to her children (see Bleidorn, Buyukcan-Tetik, van Scheppingen, Denissen, & Finkenauer, 2016). This might, for example, be due to a less strong adjustment of her achievement-related aspirations, less strong contingencies between her child's actions and positive affect, and a stronger decrease in perceived goal-attainment, as compared to other mothers.

Conclusions and implications

On a macro-analytical level, empirical research on personality development has revealed rich and well-established findings regarding developmental patterns of traits across the lifespan, and rich insights regarding classes of developmental precursors ranging from the joint influence of environmental and genetic factors to more specific effects related to life events and social roles. To further unravel the mechanisms of personality development on a micro-analytical level, empirical research needs to address and incorporate the analysis of state processes underlying these patterns and effects.

The present article aimed at introducing a first version of a model that aims to unravel these mechanisms by combining three theoretical ingredients: (1) the sequences of specific state processes that drive trait development, most comprehensively described in the TESSERA model (Wrzus & Roberts, 2016); (2) the sorting of state processes into pre-action, action, and post-action domains, following behavior-/self-regulatory models (Carver & Scheier, 1981; Wood & Denissen, 2015); and (3) the deconstruction of latent traits existing as separate entities independent of personality states, and conceptualizing trait development as state network development in the search of an equilibrium (e.g. Cramer et al., 2012). The last two points differentiate the present model from the already more elaborated TESSERA model. In our model, we do not conceptualize changes in state sequences as mediating changes in separate trait entities; instead, changes in state sequences within and across state domains are conceptualized as representing trait change. Consequently, we also describe domains of personality exclusively as state dynamics, thereby covering the whole range of individual differences sorted into domains of states of the self-regulatory cycle. This purely state and process-based representation of all kinds of personality traits has several closely related implications for the conceptualization of personality development and its empirical analysis.

First, it urges the researcher to define a priori in much more specific terms what is meant with a personality trait and with its development. Previous research typically defined trait development by stability and change in latent trait entities, which were assessed by established proxy measures independent of underlying state dynamics. The deconstruction of latent traits in the current model means that latent trait development is captured within the relevant state expressions and state contingencies. They constitute the state network that defines the trait and the relevant changes that define different trait levels. Using this perspective, defining traits and their development can be based on both theoretical reasoning (i.e., using conceptual definitions of traits and the processes that make up traits) and based on empirical analyses (i.e., investigations of the connections between states over time, such as analyses on what kind of states cluster and on the different levels of equilibrium these state clusters can reach).

Second, our model might help to clarify the interpretation of standard assessments of personality traits (e.g. global self- and informant reports of personality). Strictly speaking, latent traits make some sense from a diagnostic/prediction perspective but less so from a theoretical/explanation perspective (as they do not reflect circumscribed conceptual entities or causal units of personality; also see Asendorpf, 2016; Franić, Borsboom, Dolan, & Boomsma, 2014; Schmittmann et al., 2013). This being said, our model can also be used to somewhat align such established proxy measures of personality to relevant state domains (e.g. personality self- and informant-reports as well as implicit self-concept measures to the Evaluation and reflections domain; behavioral, interpersonal perception, cognitive, and affective measures in circumscribed laboratory contexts to the Actions and Experiences domain; explicit and implicit measures of motives and goals to the Goals and Strategies domain). Such an alignment might help to better understand what observed trait changes based on such measures really reflect.

Third, it directly implies recommendations for the development and use of alternative state-based assessments of personality. Optimally, regarding all three state domains one would need to try to assess relevant states as they are happening and continuously. The increasing use and development of experience-sampling, smartphone-sensing, ambulatory assessment, and Big data techniques provide an incredibly rich endeavor for future research on personality development—at the same time, however, it is technically, analytically, and ethically challenging (see Geukes, van Zalk, & Back, in press; Harari et al., in press; Wrzus & Mehl, 2015; for methodological overviews). This means that personality development analyses over longer periods of time would not necessarily need to include proxy measures of personality traits, but could be based on the repeated or even continuous assessment of circumscribed states. Personality trait stability and change would be analyzed as the stability and change in individual state network equilibria within and across state domains.

Fourth, the state process model represents a more parsimonious (as it does not need the assumption of latent traits as entities in addition to states) and more specific (as it describes traits in terms of concrete state mechanisms) understanding of trait development. It may, therefore, be used to derive less fuzzy, and even mathematically formalized theories based on the strengths and connections of specified state variables within and across state domains. We hope that this will allow for the testing of a rich variety of more parsimonious models and specified hypotheses in the future.

A very general type of prediction can be made regarding the kind of state processes related to the development of different kinds of traits. Essentially this directly follows from the state-based definition of traits. Some traits are broader and defined by various state-processes that even span across state domains. Conscientiousness, for example, is currently defined by states that cut across all three state domains (e.g. high valuation of work, working persistently, hard-working self-concept and reputation). Consequently, changes in these traits would be reflected in changes in all of these

various state processes. In contrast, other traits are defined by a relatively circumscribed set of state processes within state domains. Self-esteem, for example, pertains to a limited set of Evaluations and Reflections, whereas achievement motivation, in contrast, pertains to a limited set of states in the Goals and Strategies domain. Consequently, changes in these traits are reflected in this more selective set of state changes.

Following our model, one would furthermore hypothesize macro-level determinants of trait development to exert their influence via distinct state domains, with environmental and biological structures first affecting the Actions and Experiences domain and social roles/developmental tasks first affecting the Evaluations and Reflections as well as the Goals and Strategies domain. It would be highly interesting to adopt this perspective for empirical analyses on the development of broad traits that cut across process domains, thereby showing how state dynamics change within one domain (acquiring a new equilibrium within one domain), and then subsequently change the other domains (leading to an equilibrium across state domains). Many other and highly specific predictions can be tested once our generic model is used to derive more circumscribed theories.

Funding

The authors received no financial support for the research, authorship, and/or publication of this article.

Notes

1. Please note that, following dual process perspectives in personality psychology (Back & Nestler, in press), all of the described state processes can manifest on more controlled/explicit/reflective or more automatic/implicit/impulsive levels. The TESSERA model explicitly makes this distinction for reflective and associative transformation processes as well as for explicit and implicit personality self-concept representations. In our view, similar dual-process distinctions can be made for the remaining state processes, including the perception of environmental triggers, the expression of affect and behavior, and the evoked reactions to motivational underpinnings.
2. Please note that the described model can be used for person- and variable centered approaches to personality development. Following a person-centered approach, each individual can be described by a large range of state levels and contingencies that can be conceptualized as multiple interrelated state-networks. Each of these state networks can be seen as reflecting a particular trait, thus, they together form an individual's personality profile. Longitudinal observations, on all relevant state levels and contingencies of the state networks can be used to explain each individuals profile development. Following a variable-centered approach, one would focus on one state network representing one particular trait (i.e., "variable") and then use observations of multiple individuals on all relevant state levels and contingencies of this network to describe and explain shared (i.e., normative) and unique (i.e., differential) changes in a trait.

References

Asendorpf, J. B. (2016). Causal unity of broader traits is an illusion. *European Journal of Personality*, *30*, 304–305.

Back, M. D., Baumert, A., Denissen, J. J. A., Hartung, F.-M., Penke, L., Schmukle, S. C., . . . Wrzus, C. (2011). PERSOC: A unified framework for understanding the dynamic interplay of personality and social relationships. *European Journal of Personality*, *25*, 90–107.

Back, M. D., & Nestler, S. (in press). Dual process approaches to personality. In R. Deutsch, B. Gawronski & W. Hofmann (Eds.), *Reflective and impulsive determinants of human behavior*. New York, NY: Psychology Press.

Bauer, J. J., & McAdams, D. P. (2004). Personal growth in adults' stories of life transitions. *Journal of Personality*, *72*, 573–602.

Bleidorn, W. (2012). Hitting the road to adulthood: Short-term personality development during a major life transition. *Personality and Social Psychology Bulletin*, *38*, 1594–1608.

Bleidorn, W. (2015). What accounts for personality maturation in early adulthood? *Current Directions in Psychological Science*, *24*, 245–253.

Bleidorn, W., Buyukcan-Tetik, A., van Scheppingen, M. A., Denissen, J. J. A., & Finkenauer, C. (2016). Stability and change in self-esteem during the transition to parenthood. *Social Psychological and Personality Science*, *7*, 560–569. doi:10.1177/1948550616646428

Bleidorn, W., Kandler, C., & Caspi, A. (2014). The behavioral genetics of personality development in adulthood—Classic, modern, and future trends. *European Journal of Personality*, *28*, 244–255.

Bringmann, L. F., Pe, M. L., Vissers, N., Ceulemans, E., Borsboom, D., Vanpaemel, W., & Kuppens, P. (2016). Assessing emotional dynamics using networks. *Assessment*, *23*, 425–435. doi:10.1177/1073191116645909

Carver, C. S., & Scheier, M. E. (1981). *Attention and self-regulation: A control theory approach to human behavior*. New York, NY: Springer.

Caspi, A. (1987). Personality in the life course. *Journal of Personality and Social Psychology*, *53*, 1203–1213.

Caspi, A., & Roberts, B. W. (1999). Personality continuity and change across the life course. In L. A. Pervin & O. P. John (Eds.), *Handbook of personality. Theory and research* (2nd ed., pp. 300–346). New York, NY: Guilford Press.

Cervone, D. (2005). Personality architecture: Within-person structures and processes. *Annual Review of Psychology*, *56*, 423–452.

Cramer, A. O. J., van der Sluis, S., Noordhof, A., Wichers, M., Geschwind, N., Aggen, S. H., . . . Borsboom, D. (2012). Dimensions of normal personality as networks in search of equilibrium: you can't like parties if you don't like people. *European Journal of Personality*, *26*, 414–431.

Crick, N., & Dodge, K. (1994). A review and reformulation of social information-processing mechanisms in children's social adjustment. *Psychological Bulletin*, *115*, 74–101.

Denissen, J. J. A., & Penke, L. (2008). Motivational individual reaction norms underlying the five-factor model of personality: First steps towards a theory-based conceptual framework. *Journal of Research in Personality*, *42*, 1285–1302.

Denissen, J. J. A., Ulfers, H., Lüdtke, O., Muck, P. M., & Gerstorf, D. (2014). Longitudinal transactions between personality and occupational roles: A large and heterogeneous study of job beginners, stayers, and changers. *Developmental Psychology*, *50*, 1931–1942.

Denissen, J. J. A., Wood, D., Penke, L., & van Aken, M. A. G. (2013). Self-regulation underlies temperament and personality: An integrative developmental framework. *Child Development Perspectives*, *7*, 255–260.

Donnellan, M. B., Kenny, D. A., Trzesniewski, K. H., Lucas, R. E., & Conger, R. D. (2012). Using trait-state models to evaluate the longitudinal consistency of global self-esteem from adolescence to adulthood. *Journal of Research in Personality*, *46*, 634–645.

Dunlop, W. L. (2015). Contextualized personality, beyond traits. *European Journal of Personality*, *29*, 310–325.

Erikson, E. H. (1950). *Childhood and society*. New York, NY: Norton.

Feather, N. T. (1959). Subjective probability and decision under uncertainty. *Psychological Review*, *66*, 150–164.

Fleeson, W., & Jayawickreme, E. (2015). Whole trait theory. *Journal of Research in Personality*, *56*, 82–92.

Franić, S., Borsboom, D., Dolan, C. V., & Boomsma, D. I. (2014). The big five personality traits: Psychological entities or statistical constructs? *Behavior genetics*, *44*, 591–604.

Geukes, K., van Zalk, M. H. W., & Back, M. D. (in press). Analyzing processes in personality development. In J. Specht (Ed.). *Personality development across the lifespan*. San Diego, CA: Elsevier.

Gollwitzer, P. M. (1990). Action phases and mind-sets. In E. T. Higgins & R. M. Sorrentino (Eds.), *Handbook of motivation and cognition: Foundations of social behavior* (Vol. *2*, pp. 53–92). New York, NY: Guilford Press.

Gross, J. J., & Thompson, R. A. (2007). Emotion regulation: Conceptual foundations. In J. J. Gross (Ed.), *Handbook of emotion regulation* (pp. 3–24). New York, NY: Guilford Press.

Harari, G. M., Lane, N. D., Wang, R., Crosier, B. S., Campbell, A. T., & Gosling, S. D. (in press). Using smartphones to collect behavioral data in psychological science: Opportunities, practical considerations, and challenges. *Perspectives on Psychological Science*.

Hastie, R. (2001). Problems for judgment and decision making. *Annual Review of Psychology*, *52*, 653–683.

Havighurst, R. J. (1972). *Developmental tasks and education*. New York, NY: McKay Company.

Heckhausen, H. (1991). *Motivation and action*. New York, NY: Springer-Verlag.

Headey, B., & Wearing, A. (1989). Personality, life events, and subjective well-being: Toward a dynamic equilibrium model. *Journal of Personality and Social Psychology*, *57*, 731–739.

Hennecke, M., Bleidorn, W., Denissen, J. J.A., & Wood, D. (2014). A three-part framework for self-regulated personality development across adulthood. *European Journal of Personality*, *28*, 289–299.

Higgins, E. T. (1987). Self-discrepancy: A theory relating self and affect. *Psychological Review*, *94*, 319–340.

Hutteman, R., Bleidorn, W., Keresteš, G., Brković, I., Butković, A., & Denissen, J. J. A. (2014). Reciprocal associations between parenting challenges and parents' personality development in young and middle adulthood. *European Journal of Personality*, *28*, 168–179.

Hutteman, R., Hennecke, M., Orth, U., Reitz, A. K., & Specht, J. (2014). Developmental tasks as a framework to study personality development in adulthood and old age. *European Journal of Personality*, *28*, 267–278.

Hutteman, R., Nestler, S., Wagner, J., Egloff, B., & Back, M. D. (2015). Wherever I may roam: Processes of self-esteem development from adolescence to emerging adulthood in the context of international student exchange. *Journal of Personality and Social Psychology*, *108*, 767–783.

Lodi-Smith, J., Geise, A. C., Roberts, B. W., & Robins, R. W. (2009). Narrating personality change. *Journal of Personality and Social Psychology*, *96*, 679–689.

Lodi-Smith, J., & Roberts, B. W. (2007). Social investment and personality: A meta-analysis of the relationship of personality traits to investment in work, family, religion, and volunteerism. *Personality and Social Psychology Review*, *11*, 68–86.

Lüdtke, O., Roberts, B. W., Trautwein, U., & Nagy, G. (2011). A random walk down university avenue: Life paths, life events, and personality trait change at the transition to university life. *Journal of Personality and Social Psychology*, *101*, 620–637.

Mischel, W., & Shoda, Y. (1998). Reconciling processing dynamics and personality dispositions. *Annual Review of Psychology*, *49*, 229–258.

Nickerson, R. S. (1998). Confirmation bias: A ubiquitous phenomenon in many guises. *Review of General Psychology*, *2*, 175–220.

Ormel, J., Riese, H., & Rosmalen, J. G. M. (2012). Interpreting neuroticism scores across the adult life course: Immutable or experience-dependent set points of negative affect? *Clinical Psychology Review*, *32*, 71–79.

Orth, U., & Robins, R. W. (2014). The development of self-esteem. *Current Direction in Psychological Science*, *23*, 381–387.

Orth, U., Trzesniewski, K. H., & Robins, R. W. (2010). Self-esteem development from young adulthood to old age: A cohort-sequential longitudinal study. *Journal of Personality and Social Psychology*, *98*, 645–658.

Pasupathi, M., Mansour, E., & Brubaker, J. R. (2007). Developing a life story: Constructing relations between self and experience in autobiographical narratives. *Human Development*, *50*, 85–110.

Reitz, A. K., Motti-Stefanidi, F., & Asendorpf, J. B. (2013). Mastering developmental transitions in immigrant adolescents: The longitudinal interplay of family functioning, developmental and acculturative tasks. *Developmental Psychology*, *50*, 754–765.

Roberts, B. W. (2006). Personality development and organizational behavior. In B. M. Staw (Ed.), *Research on Organizational Behavior* (Vol. *27*, pp. 1–40). New York, NY: Elsevier Science/JAI Press.

Roberts, B. W. (2009). Back to the future: Personality and assessment and personality development. *Journal of Research in Personality*, *43*, 137–145.

Roberts, B. W., Donnellan, M. B., & Hill, P. L. (2012). Personality trait development in adulthood: Findings and implications. In H. Tennen & J. Suls (Eds.), *Handbook of psychology* (2nd ed., pp. 183–196). Hoboken, NJ: Wiley.

Roberts, B. W., & Jackson, J. J. (2008). Sociogenomic personality psychology. *Journal of Personality*, *76*, 1523–1544.

Roberts, B. W., Walton, K., & Viechtbauer, W. (2006). Patterns of mean-level change in personality traits across the life course: A meta-analysis of longitudinal studies. *Psychological Bulletin*, *132*, 1–25.

Roberts, B. W., & Wood, D. (2006). Personality development in the context of the neo-socioanalytic model of personality. In D. K. Mroczek & T. D. Little (Eds.), *Handbook of personality development* (pp. 11–39). Mahwah, NJ: Lawrence Erlbaum Associates.

Roberts, B. W., Wood, D., & Caspi, A. (2008). The development of personality traits in adulthood. In O. P. John, R. W. Robins & L. A. Pervin (Eds.), *Handbook of personality: Theory and research* (3rd ed., pp. 375–398). New York, NY: Guilford Press.

Robins, R. W., & Trzesniewski, K. H. (2005). Self-esteem development across the lifespan. *Current Directions in Psychological Science*, *14*, 158–162.

Robinson, M. D. (2007). Personality, affective processing, and self-regulation: Toward process-based views of extraversion, neuroticism, and agreeableness. *Social and Personality Psychology Compass*, *1*, 223–235.

Robinson, M. D., & Wilkowski, B. M. (2015). Personality processes and processes as personality: A cognitive perspective. In: M. Mikulincer & P. R. Shaver (Eds.), *APA handbook of personality and*

social psychology (Vol. *4*, pp. 129–145). Washington, DC: American Psychological Association.

Rotter, J. B. (1954). *Social learning and clinical psychology*. New York, NY: Prentice-Hall.

Schmittmann, V. D., Cramer, A. O., Waldorp, L. J., Epskamp, S., Kievit, R. A., & Borsboom, D. (2013). Deconstructing the construct: A network perspective on psychological phenomena. *New Ideas in Psychology*, *31*, 43–53.

Specht, J., Bleidorn, W., Denissen, J. J., Hennecke, M., Hutteman, R., Kandler, C., . . . Zimmermann, J. (2014). What drives adult personality development? A comparison of theories and empirical evidence. *European Journal of Personality*, *28*, 216–230.

Specht, J., Egloff, B., & Schmukle, S. C. (2011). Stability and change of personality across the life course: The impact of age and major life events on mean-level and rank-order stability of the Big Five. *Journal of Personality and Social Psychology*, *101*, 862–882.

Srivastava, S., John, O. P., Gosling, S. D., & Potter, J. (2003). Development of personality in early and middle adulthood: Set like plaster or persistent change? *Journal of Personality and Social Psychology*, *84*, 1041–1053.

Trzesniewski, K. H., Donnellan, M. B., & Robins, R. W. (2003). Stability of self-esteem across the life span. *Journal of Personality and Social Psychology*, *84*, 205–220.

Wagner, J., Gerstorf, D., Hoppmann, C., & Luszcz, M. (2013). The nature and correlates of self-esteem trajectories in late life. *Journal of Personality and Social Psychology*, *105*, 139–153.

Wagner, J., Lüdtke, O., Jonkmann, K., & Trautwein, U. (2013). Cherish yourself: Longitudinal patterns and conditions of self-esteem change in the transition to young adulthood. *Journal of Personality and Social Psychology*, *104*, 148–163.

Wagner, J., Lüdtke, O., & Trautwein, U. (2015). Self-esteem is mostly stable across young adulthood: Evidence from latent STARTS models. *Journal of Personality*, *84*, 523–535.

Wood, D., & Denissen, J. J. A. (2015). A functional perspective on personality trait development. In N. R. Branscombe & K. Reynolds (Eds.), *Psychology of change: Life contexts, experiences, and identities*. New York, NY: Psychology Press.

Wood, D., Gardner, M. H., & Harms, P. D. (2015). How functionalist and process approaches to behavior can explain trait covariation. *Psychological Review*, *122*, 84–111.

Wright, A. G. C., & Hopwood, C. J. (2016). Advancing the assessment of dynamic psychological processes. *Assessment*, *23*, 399–403. doi: 10.1177/1073191116654760.

Wrzus, C., & Mehl, M. (2015). Lab and/or field? Measuring personality processes and their social consequences. *European Journal of Personality*, *29*, 250–271.

Wrzus, C., & Roberts, B. W. (2016). Processes of personality development in adulthood: The TESSERA framework. *Personality and Social Psychology Review*. Advance online publication. doi:10.1177/1088868316652279

Zimmermann, J., & Neyer, F. J. (2013). Do we become a different person when hitting the road? Personality development of sojourners. *Journal of Personality and Social Psychology*, *105*, 515–530.

View publication stats